

A systematic large-cap equity mandate balancing three stock-selection criteria

Investor need

Systematic large-cap stock selection that does **not** make one characteristic the sole driver of the book — while remaining fully invested in equities and keeping sector weights close to the eligible universe.

Defined product

- Long-only U.S. large-cap stocks
- Equal-weight composite of valuation, profitability, and momentum ranks within sector
- Sector targets matched to the eligible universe at rebalance
- Monthly rebalance · next-session-close execution

Conditional ask

Illustrative \$10 million AUM for this stock-level mandate — if point-in-time validation and operational readiness clear.

Not working capital. Current exploratory research does **not** establish readiness to deploy.

Hypothesis: combining different information may reduce dependence on any single selection criterion.

RATIONALE

Why combine? An economic rationale for combining signals

Valuation

$$V = B / M$$

Book equity over contemporaneous market cap.

Profitability proxy

$$Q = NI^{\text{ann}} / \bar{B}$$

Annual net income over period-matched average book — **not** full balance-sheet strength.

Momentum

$$M = P_{t-21} / P_{t-252} - 1$$

Trading-session 12–1 total return.

- *Value and Momentum Everywhere* (Asness, Moskowitz & Pedersen, Journal of Finance, 2013) documents diversification benefits from combining value and momentum across markets — a reason to investigate combinations beyond a short-sample winner. It does **not** validate this long-only ranking formula. Source: aqr.com / Journal of Finance.
- Adding signals ≠ reducing risk with cash: value and profitability change **which businesses** you own while staying fully invested.
- Composite $S_i = (V_i + Q_i + M_i) / 3$ balances overall scores; a name can offset weak profitability with strong value and momentum.
- Equal weights: transparent starting choice, not fit to this short sample.

IMPLEMENTATION

Exact rules of the stock mandate

Construction

- ~Top 500 by market cap; price $\geq$ \$5; full 20-session ADV $\geq$ \$1mm
- Sector percentile ranks $\rightarrow$ equal-weight composite $\rightarrow$ top ~20% / sector
- Sector weights = eligible-universe cap weights at rebalance
- Signal at month-end close $\rightarrow$ trade at **next session close**

Design notes

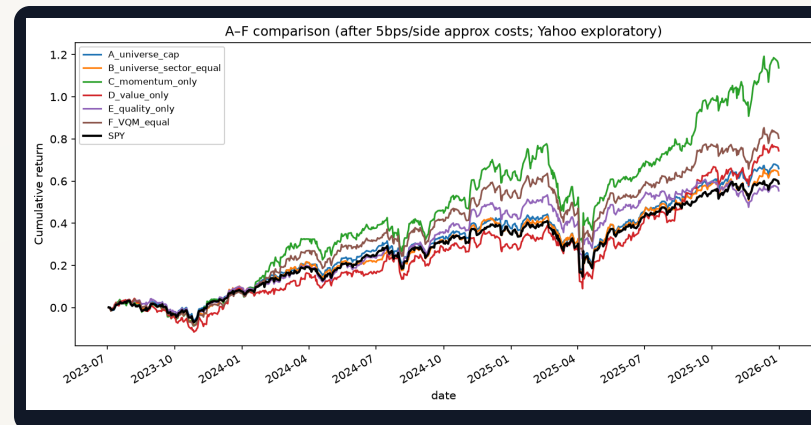
- We balance valuation, profitability, and momentum in selection — we do **not** claim every holding is cheap, high-profitability, and trending.
- Sector controls also apply to single-factor comparators in our tests — they do not uniquely distinguish VQM from momentum-only.
- Analysis uses estimated trading costs of 5 bps per side (proportional approximation).

Sample trade-offs versus simpler alternatives

After estimated trading costs of 5 bps per side; before management fees (unless noted).

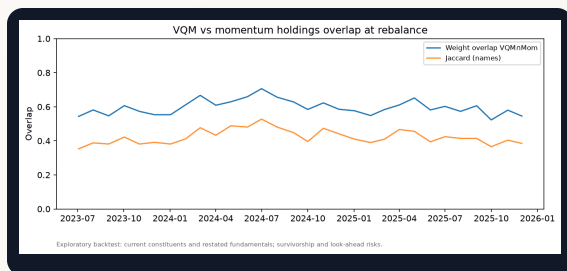
PORTFOLIO	CAGR	VOL	SHARPE	MAX DD
VQM	26.71%	18.04%	1.13	-22.87%
Momentum-only	35.64%	21.18%	1.32	-26.36%
Universe cap-wt	22.54%	15.39%	1.08	-19.49%
SPY	20.40%	15.51%	0.96	-18.76%

- Vs momentum: **lower observed volatility and maximum drawdown**; lower return and Sharpe.
- Vs universe: higher return; higher vol and deeper DD; modest Sharpe lift.
- Average one-way turnover: **27.63%** including initial formation (engine summary); **25.48%** excluding initial formation (29 subsequent rebalances).

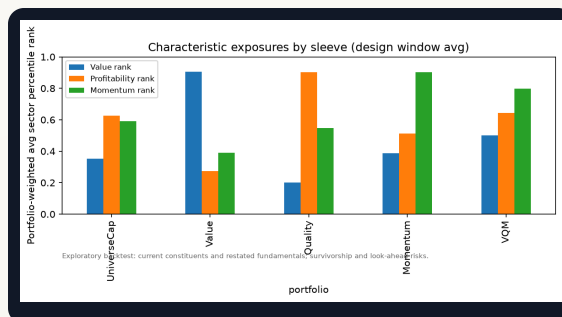


COMPLEMENTARITY CHECKS

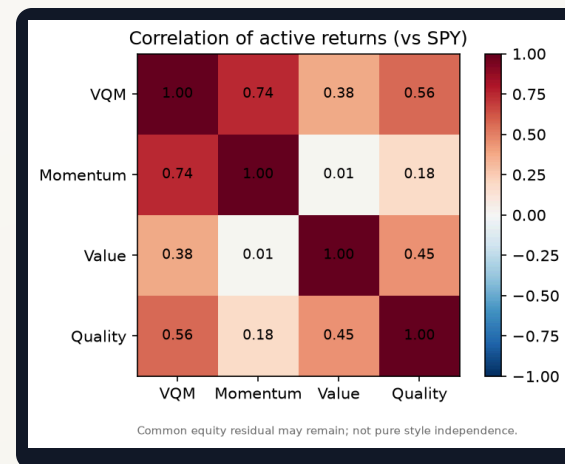
Do holdings and exposures actually differ?



Averages across **30** rebalances: VQM vs Mom Jaccard **42.3%**, weight overlap **59.7%**. Mom vs Value weight overlap **9.9%**.



Portfolio-weighted average sector percentile ranks for VQM: value **0.50** / profitability **0.64** / momentum **0.80** — **momentum-tilted with additional profitability exposure**; value near the middle of the ranking universe (~0.50).



Correlations of **active returns** (sleeve return – SPY daily return). Value–Mom ≈ 0.01 ; VQM–Mom ≈ 0.74 . Not regression residuals.

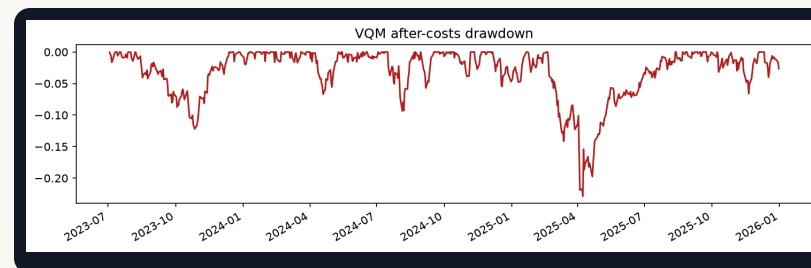
Three labels do not guarantee three independent return sources. Single sleeves differ materially; the composite still leans on momentum in this sample.

What remains open — and \$10mm trading

Limitations

- Short sample; current constituents; restated fundamentals; current sectors
- Profitability proxy $\neq$ comprehensive quality
- Lower vol/DD vs momentum is an observed sample feature — not an isolated momentum-crash study
- 2026 previously examined; costs are an approximation

A rationale for VQM does **not** automatically justify charging **75 bps**.
After-fee Sharpe (~ 1.09) $\approx$ universe-cap (~ 1.08) in this sample.



@ \$10MM	MEDIAN	MAX
Position / ADV	0.026%	5.595%
Rebalance trade / ADV	0.001%	5.594%

Trade/ADV excludes initial formation. Feasibility looks plausible, not proven.

CLOSE

Keep VQM — diversification of investment criteria under uncertainty

Expected questions

- *What am I funding?* The stock-level VQM mandate.
- *Why not momentum-only?* Avoiding a single selection criterion can be rational even when momentum led this window; implementation evidence is partial.
- *Ready?* Not on current evidence.
- *Sector edge vs mom sleeve?* Same sector-control design in both.

Milestones before any live allocation

1. Historical membership + as-reported fundamentals with availability dates
2. Point-in-time rebuild; re-run A-F and overlap/exposure tests
3. TCA / trade-participation pack at target AUM

Closing ask: illustrative **\$10 million** for the stock-level VQM mandate, **conditional** on those milestones. This backtest neither proves complementarity nor warrants abandoning VQM for its recent winner.

Design-window performance (≥5 decimals)

Source file: outputs/tables/AF_comparison_summary.csv · Window 2023-07-03 → 2025-12-31

STRATEGY	CAGR	ANN. VOL	SHARPE	MAX DD	TURNOVER 1-WAY	COST/FEE LAYER
F_VQM_equal	0.26712	0.18045	1.13333	-0.22866	0.27630	after_trading_cost_5bps
C_momentum_only	0.35640	0.21179	1.31642	-0.26359	0.27495	after_trading_cost_5bps
A_universe_cap	0.22539	0.15388	1.08206	-0.19487	0.02971	after_trading_cost_5bps
SPY	0.20402	0.15514	0.96085	-0.18755		ETF_ER_embedded_no_extra_mgmt
F_VQM_after_fee	0.25765	0.18044	1.09175	-0.22944	0.27630	after_trading_cost_and_75bps_mgmt
F_VQM_gross	0.27134	0.18042	1.15192	-0.22819	0.27630	gross_0bps

Turnover reconciliation

- Engine average one-way turnover (includes initial formation): **0.27630** = 27.63% (AF summary / rebalance log).
- Average across 29 subsequent rebalances only (ex-formation, from holdings weight changes): **0.25481** = 25.48%.
- One-way turnover = $\frac{1}{2} \times \sum |\Delta w_i|$ at each rebalance.

Overlap, exposures, and active-return correlations

Holdings overlap (averages across rebalances)

Each row averages Jaccard / weight overlap over n_dates rebalance dates.

PAIR	AVG JACCARD	AVG WEIGHT OVERLAP	N DATES
VQM_vs_Momentum	0.42303	0.59668	30
VQM_vs_Value	0.12276	0.22445	30
VQM_vs_Quality	0.20578	0.33554	30
Momentum_vs_Value	0.04903	0.09858	30
Momentum_vs_Quality	0.13855	0.24429	30
Value_vs_Quality	0.00945	0.01121	30

Portfolio-weighted average ranks

Characteristic ranks = portfolio-weight × sector percentile rank, averaged over rebalances.

PORTFOLIO	VALUE	PROFITABILITY	MOMENTUM
VQM	0.50225	0.64415	0.79889
Momentum	0.38682	0.51153	0.90204
Value	0.90481	0.27279	0.38897
Quality	0.20045	0.90348	0.54875
UniverseCap	0.35385	0.62628	0.59005

Active-return correlations

Active return_t = sleeve return_t – SPY return_t (same dates). Benchmark = SPY total return (Adj Close). These are **active returns**, not regression residuals.

	VQM	MOMENTUM	VALUE	QUALITY
VQM	1.00000	0.74442	0.38056	0.56473
Momentum	0.74442	1.00000	0.01273	0.18405
Value	0.38056	0.01273	1.00000	0.44819
Quality	0.56473	0.18405	0.44819	1.00000

Metric definitions

- Jaccard = $|A \cap B| / |A \cup B|$ on holding names at a rebalance.
- Weight overlap = $\sum_i \min(w_i^A, w_i^B)$.
- Ranks are within-sector percentiles among the eligible top-N universe.

Short robustness table and references

Source: outputs/tables/robustness_design.csv (design window; after-cost variants). Selected rows:

VARIANT	CAGR	SHARPE	MAX DD
selection_pct=0.1	31.05%	1.21	-23.24%
selection_pct=0.2	25.47%	1.06	-23.43%
selection_pct=0.3	22.39%	0.97	-21.56%
lag_days=45	24.67%	1.02	-23.78%
lag_days=60	25.47%	1.06	-23.43%
lag_days=90	25.51%	1.07	-23.42%
rebalance=Q	25.70%	1.06	-23.58%
cost_bps_per_side=5.0	25.47%	1.06	-23.43%
cost_bps_per_side=10.0	25.05%	1.05	-23.47%
cost_bps_per_side=15.0	24.63%	1.03	-23.52%
cost_bps_per_side=25.0	23.80%	0.99	-23.62%

Sources

- Asness, C., Moskowitz, T., & Pedersen, L. (2013). Value and Momentum Everywhere. *Journal of Finance*. Also: aqr.com research page.
- Fama, E. & French, K. (1992, 2015). Cross-section / five-factor models.
- iShares VLUE / QUAL / MTUM fact sheets (ishares.com), expense ratio 0.15% as of mid-2026 fact sheets.
- Prices/fundamentals: Yahoo Finance via yfinance (exploratory; restated). Risk-free: FRED DGS3MO. Universe panel: current S&P constituents research list.

Fee reminder (once): a rationale for combining signals does not automatically justify a 75 bps management fee.